

What is xSPI and why is it essential for MRAM?

xSPI is a flexible interface used in MRAM where 'x' can denote 1 (single), 4 (quad), or up to 8 (octal) connections. Unlike competitors who primarily use four or one connection, we offer octal SPI, enabling speeds of up to 400 megabytes per second, which is unique to us. Previously, our MRAM had a parallel interface based on industry demands. The xSPI interface reduces the number of required terminals, like the transition seen with DDR technology in other applications. This is why we transitioned from a parallel to a serial interface in newer MRAM models.

What are the key design considerations?

In designing memory products, compatibility with existing systems is prioritised. For example, when the 64-megabit STT-MRAM was introduced, compatible controllers were also provided. A special controller was developed for octal SPI, as no alternative existed at the time. Additionally, compatibility with controllers for both quad and single SPI was ensured. This system-wide approach supports the development of both memory and its ecosystem in tandem.

How does MRAM handle radiation in space?

In deep space, memory devices are exposed to alpha radiation, which can disrupt memory systems relying on electronic charges. MRAM, however, utilises electron spin instead of electronic charges, making it inherently resistant to radiation up to one million rads of silicon. This robustness is why MRAM is used in space missions, such as the Perseverance rover on Mars and the Lucy mission to Jupiter. While MRAM itself is radiation-resistant, full radiation hardening is conducted

by defence industry partners who protect the surrounding components.

What are the potential applications?

We focus on various sectors, including industrial automation, casino gaming, and storage systems. In industrial automation, we partner with major companies like Siemens and Schneider, using our advanced memory to ensure machines retain their last actions during power outages, preventing disruptions in manufacturing. In the casino gaming sector, every slot machine action must be recorded rapidly and in multiple locations, and our memory efficiently handles this compared to older technologies. For storage systems, we supply memory that stores essential data temporarily on servers, and we have captured a large share of this market. Additionally, our memory is crucial in electric vehicles (EVs) for quickly recording such data as battery conditions during acceleration, helping manufacturers monitor performance over time.

Do you think this technology can replace other storage technologies?

When plotting memory types by read-write cycles and access speed, storage memory like hard disks and SSDs would be on the bottom right of the graph, with slower access speeds and lower read-write cycles. MRAM is not designed for these applications. On the other hand, memories like DRAM and SRAM offer much faster access speeds and higher read-write cycles. Our MRAM competes with these faster technologies, achieving write cycles as fast as 35 nanoseconds. MRAM is also being considered as a replacement for NOR flash, which has slower write times (microseconds) and has stopped advancing in terms of miniaturisation past 40 nanometres.

What do you see for the future of this technology?

We are currently focusing on two main developments for the future. The first is addressing the limitations of NOR flash, which has stopped scaling at 40nm CMOS. For high-density NOR flash beyond 128 megabits, STT-MRAM offers a promising alternative in a market valued at around \$2.5 billion. We are designing a new part for this segment, with testing planned in the coming years and a target market entry around 2025-2026.

Secondly, we are exploring MRAM applications in AI inference at the edge, where it can process data locally without needing constant memory refreshing or cloud communication, resulting in significant energy savings. This non-volatile memory is ideal for reducing power consumption in edge devices. Alongside this, we aim to adapt MRAM for use in data centres, ensuring compatibility with emerging interfaces like the UCIE interface and CXL protocols. These efforts align with our goals to revolutionise AI processing at the edge and expand our presence in the storage industry.

How do you think MRAM will be used in the future of aerospace?

Aerospace equipment must be resistant to radiation, and MRAM fits this requirement well. Many of our defence industry customers in the US, and increasingly in India and Europe, use our MRAM components through partnerships with companies like Honeywell. Our technology is widely used—from ISRO in India to NASA in the US, and even under the European Space Agency. Since MRAM naturally resists radiation and performs well in extreme temperatures (ranging from -40°C to 125°C), it has strong prospects in aerospace. **EFY**